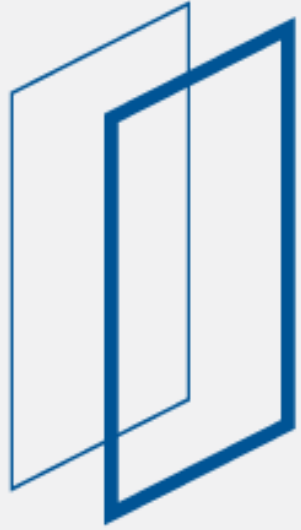




Iustitia's Oracle

**Forecasting Political Regime Types
by Measuring Judicial Independence**

Final Project by Friederike Augustin



Global Decline in Democracy

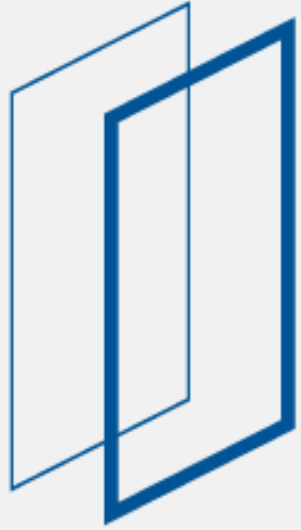
Emerging threat:

Political interference in the judiciary undermines the critical role of courts and judges in preserving democracy and upholding its core values.

Independence lies in the details:

- Who selects and appoints judges?
- How long do judges remain in office?
- Who decides on disciplinary matters?
- Who controls the budget of courts?
- etc. etc.





Project Overview

Key Question:

Can written law predict the quality of democracy?

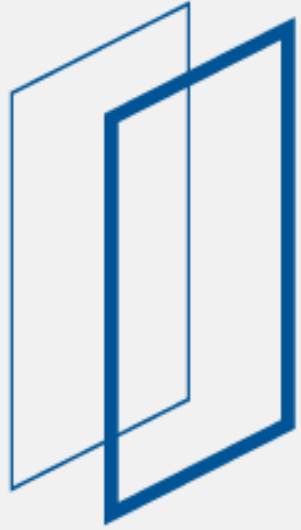
Goal:

To predict future political regime types and levels of democracy based on measures of judicial independence.

Data:

1. Dataset on political regime characteristics
2. Dataset on judicial system characteristics and reforms

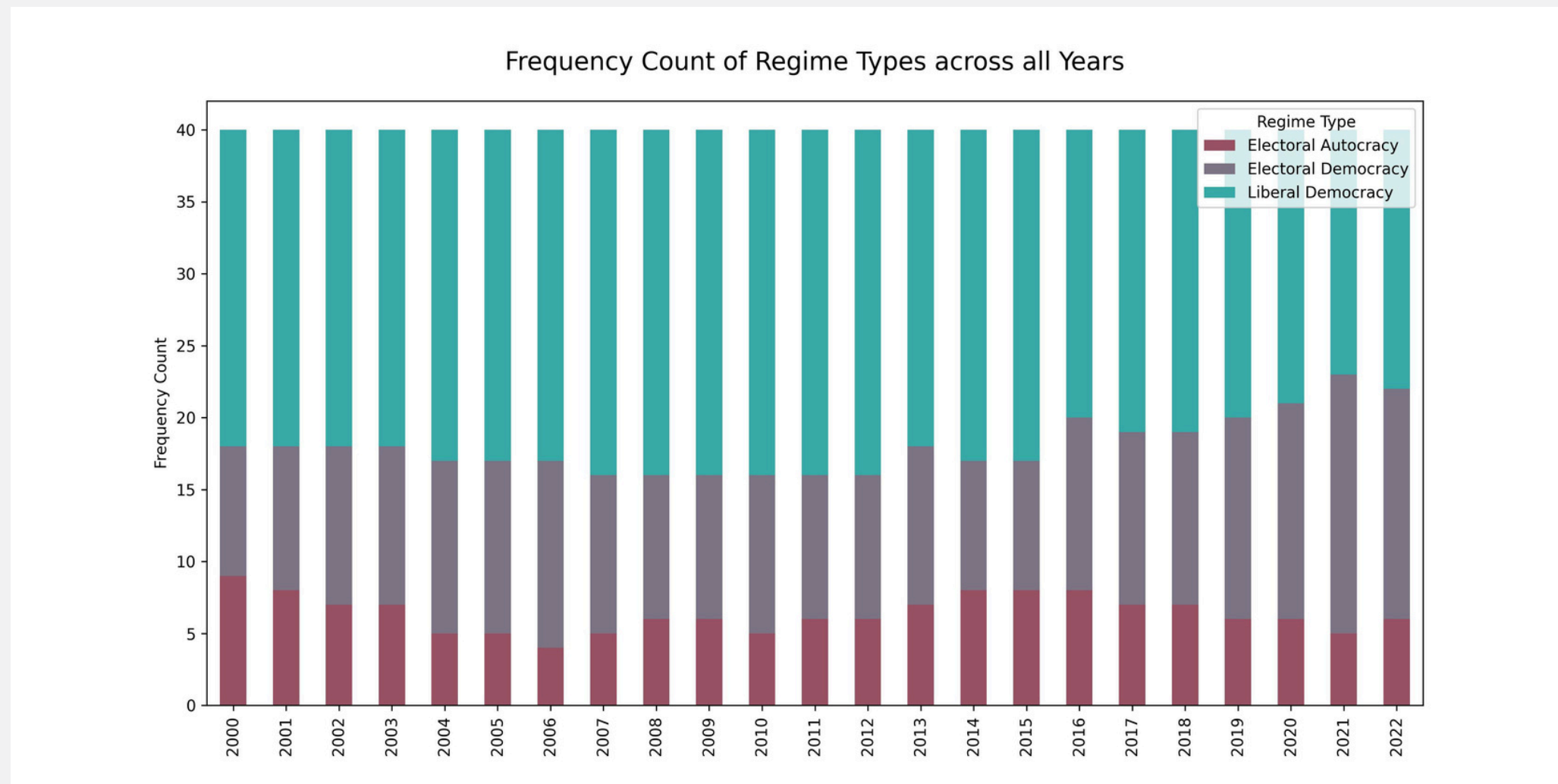


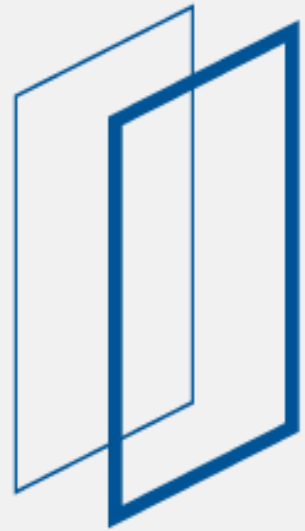


Data Selection

Varieties of Democracy Dataset (V-Dem)

- Comprehensive and detailed democracy ratings (published annually)
- Incl. democracy index [0,1] and regime types classification [0,3]

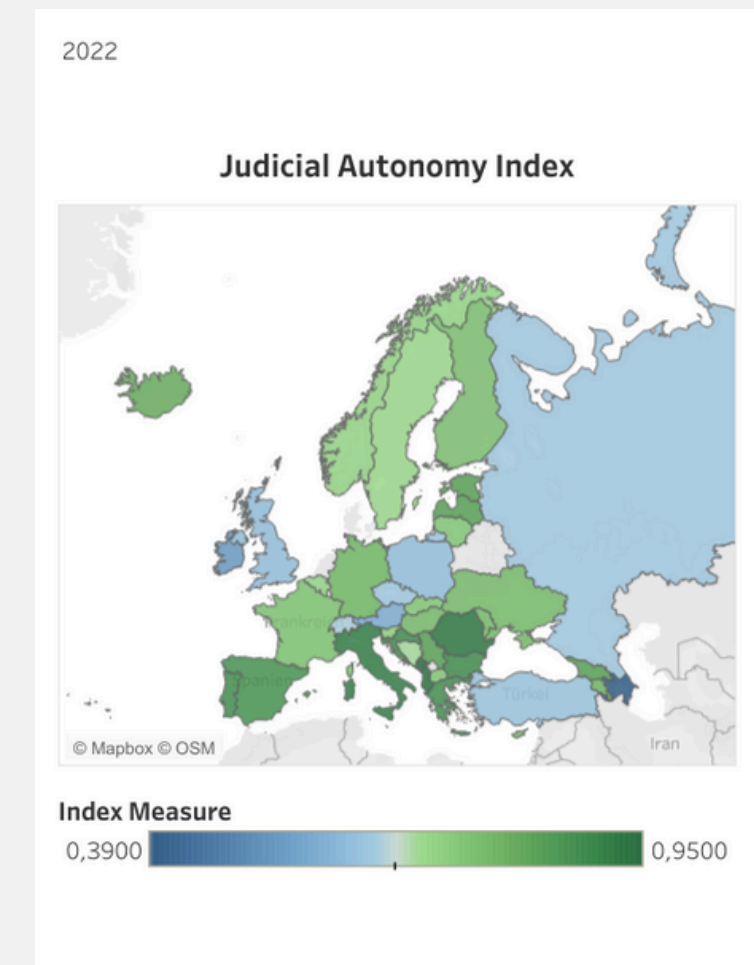
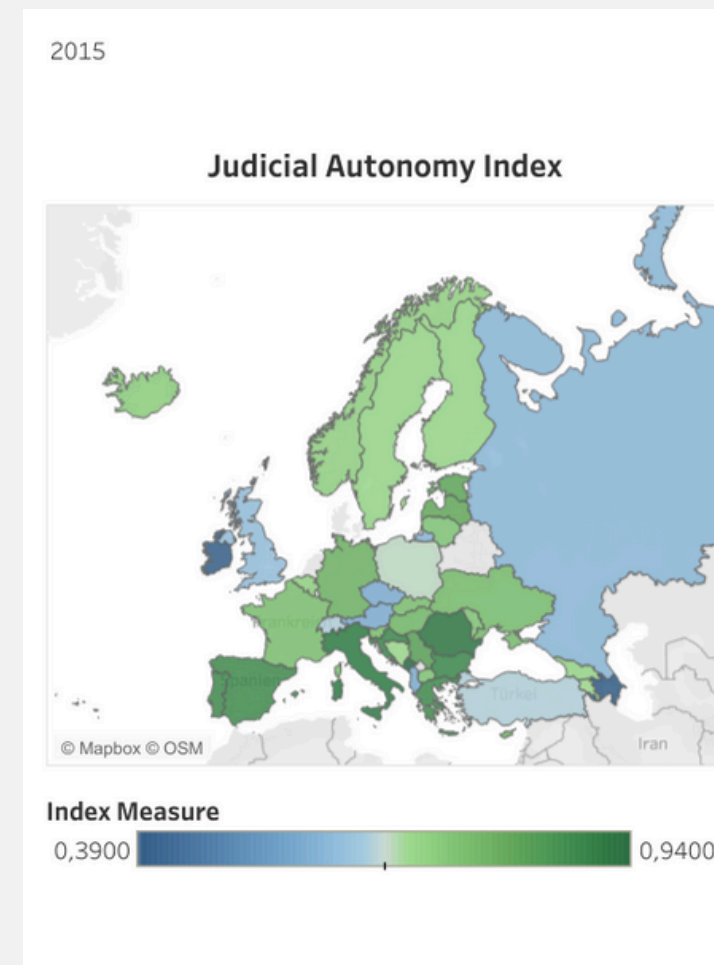
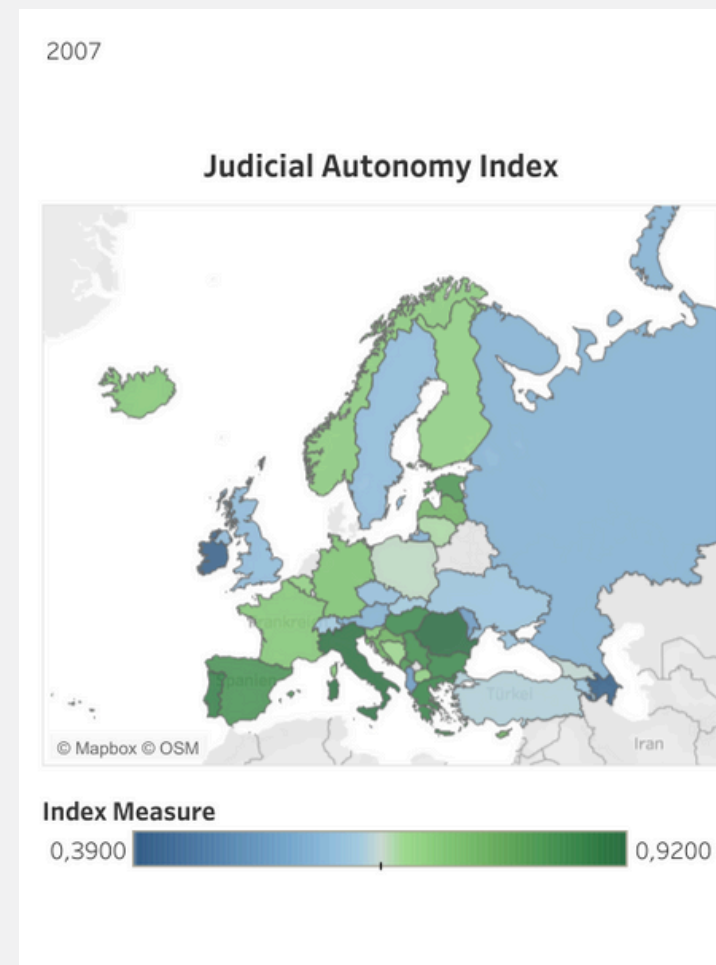
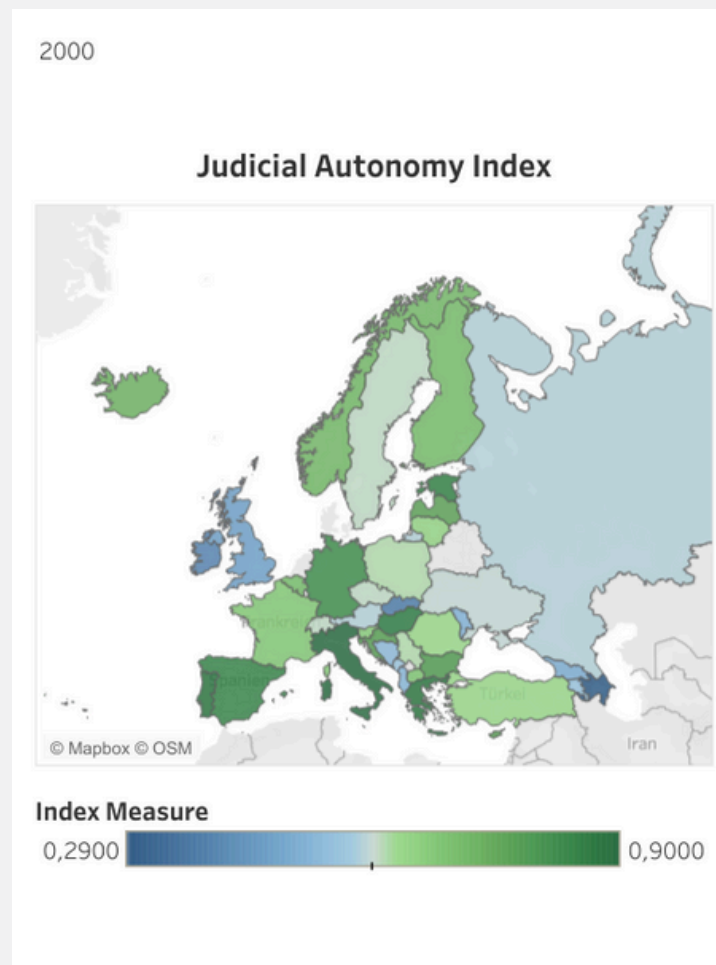


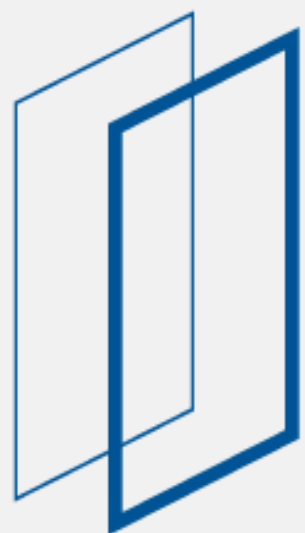


Data Selection

Judicial Autonomy Dataset (Humboldt-University Berlin)

- Are judges and courts independent from political influence?
- Coded characteristics of 40 European judicial systems for 2000-2022

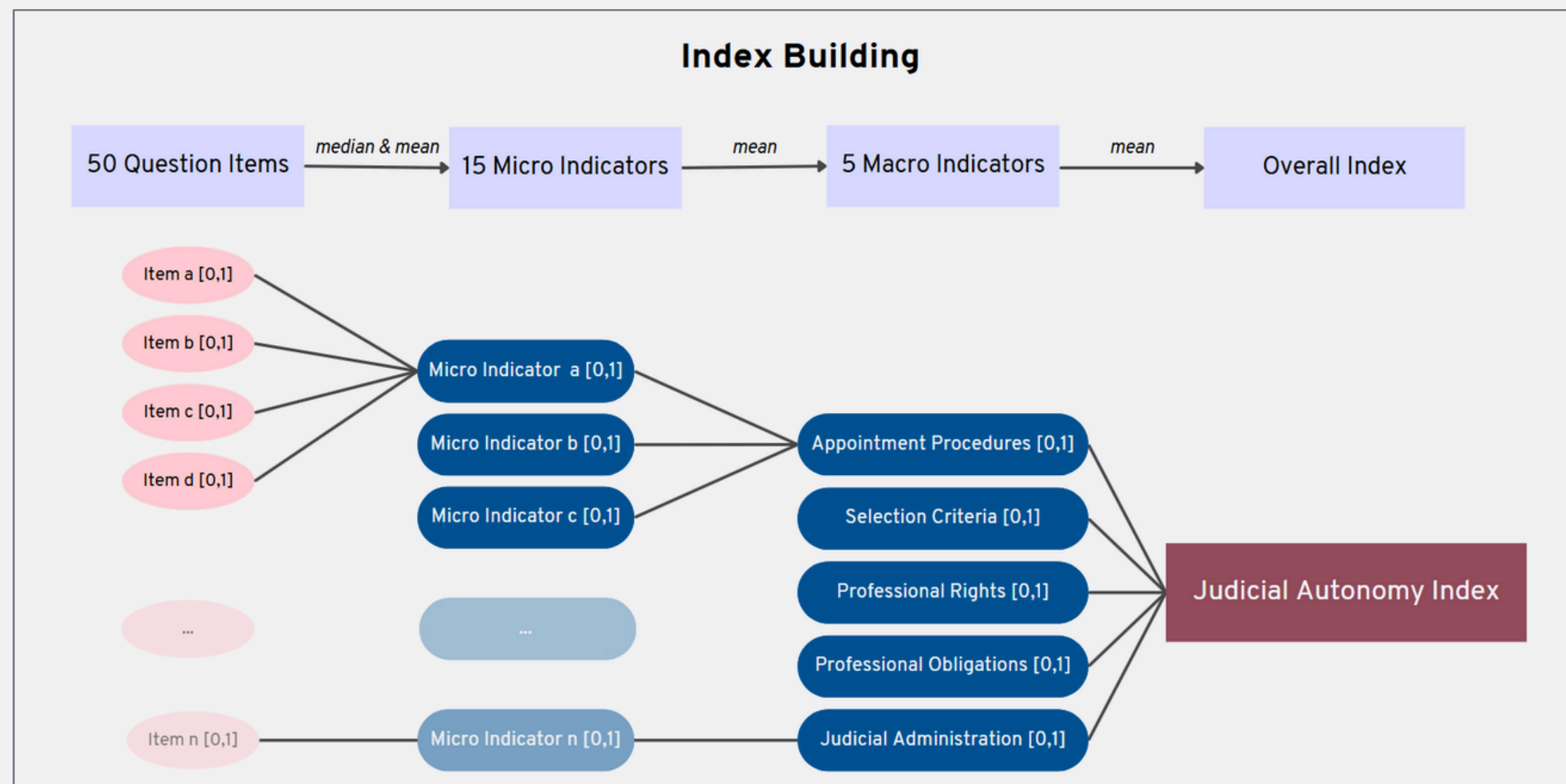


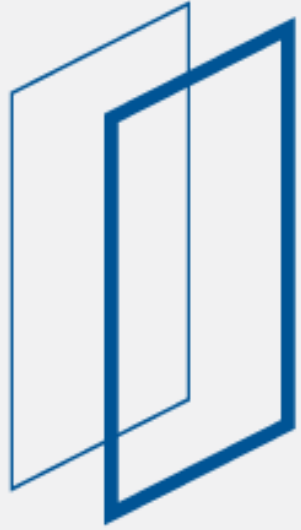


Data Preparation

Index Building:

- Filling in all missing values
- Aggregating data by mode and mean

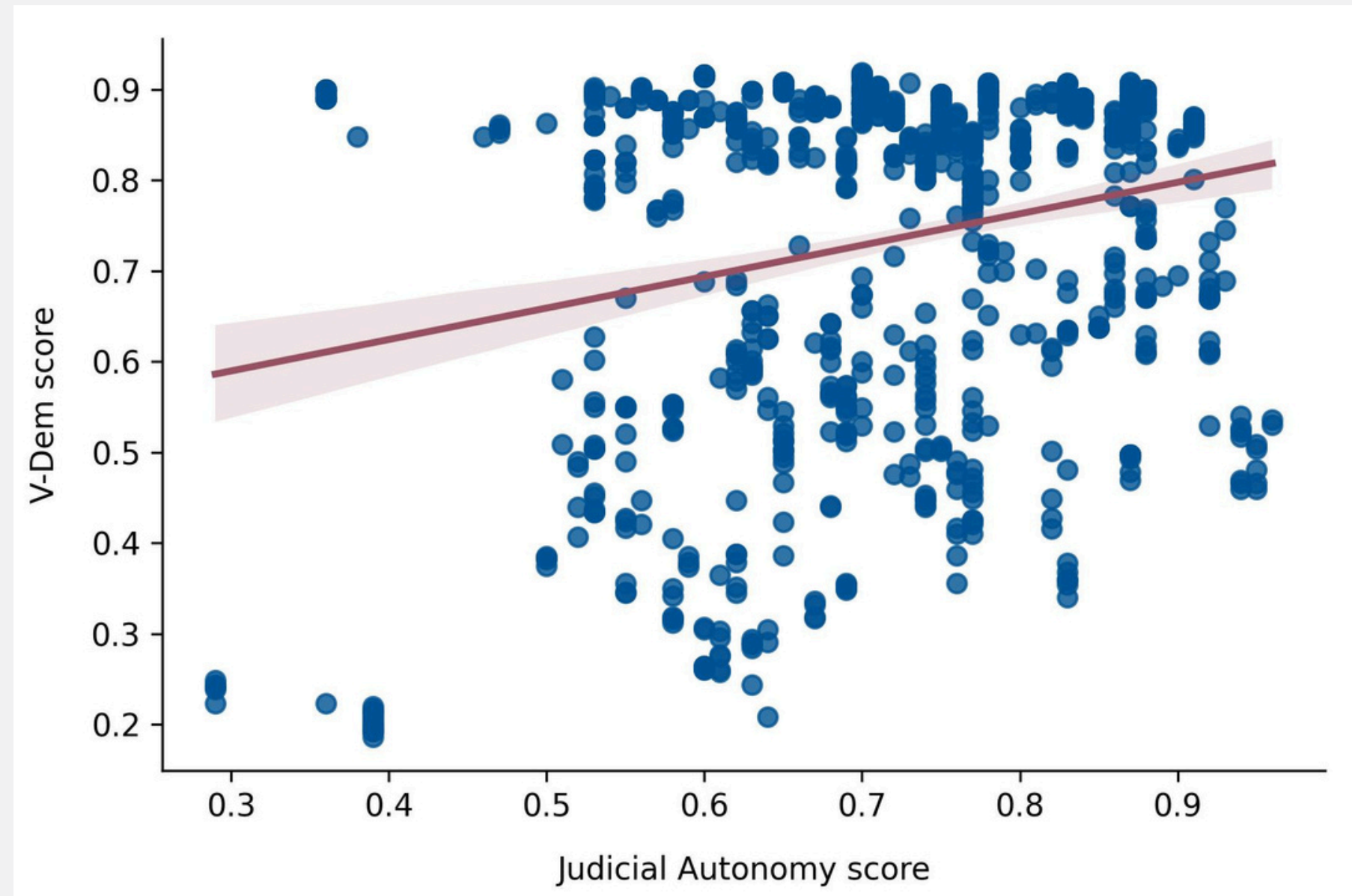


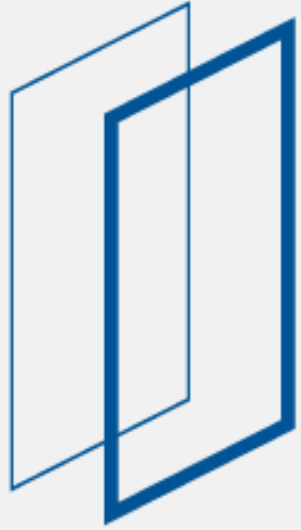


Democracy vs. Judicial Autonomy

Correlation Coefficients:

- Weak positive linear relationship (Pearson: 0.24)
- Weak positive monotonic relationship (Spearman: 0.11)





ML Model Building

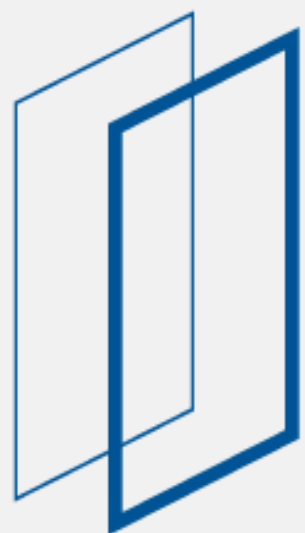
K-Nearest Neighbors

- Classification and Regression Models
- Manual feature selection, feature scaling, oversampling
- Explore best feature-target combinations (accuracy & $R^2 > 90\%$)

Random Forest

- Classification and Regression Models
- Based on pre-selected feature-target combinations
- Identify feature importances

Targets	Features	
	Judicial Autonomy Micro Indicators (15 features)	Judicial Autonomy Macro Indicators (5 features)
	V-Dem Regime Types (categorical)	V-Dem Democracy Index (numerical)
	KNNClassifier RFClassifier	KNNClassifier
	KNNRegressor RFRegressor	KNNRegressor



Random Forest Classifier

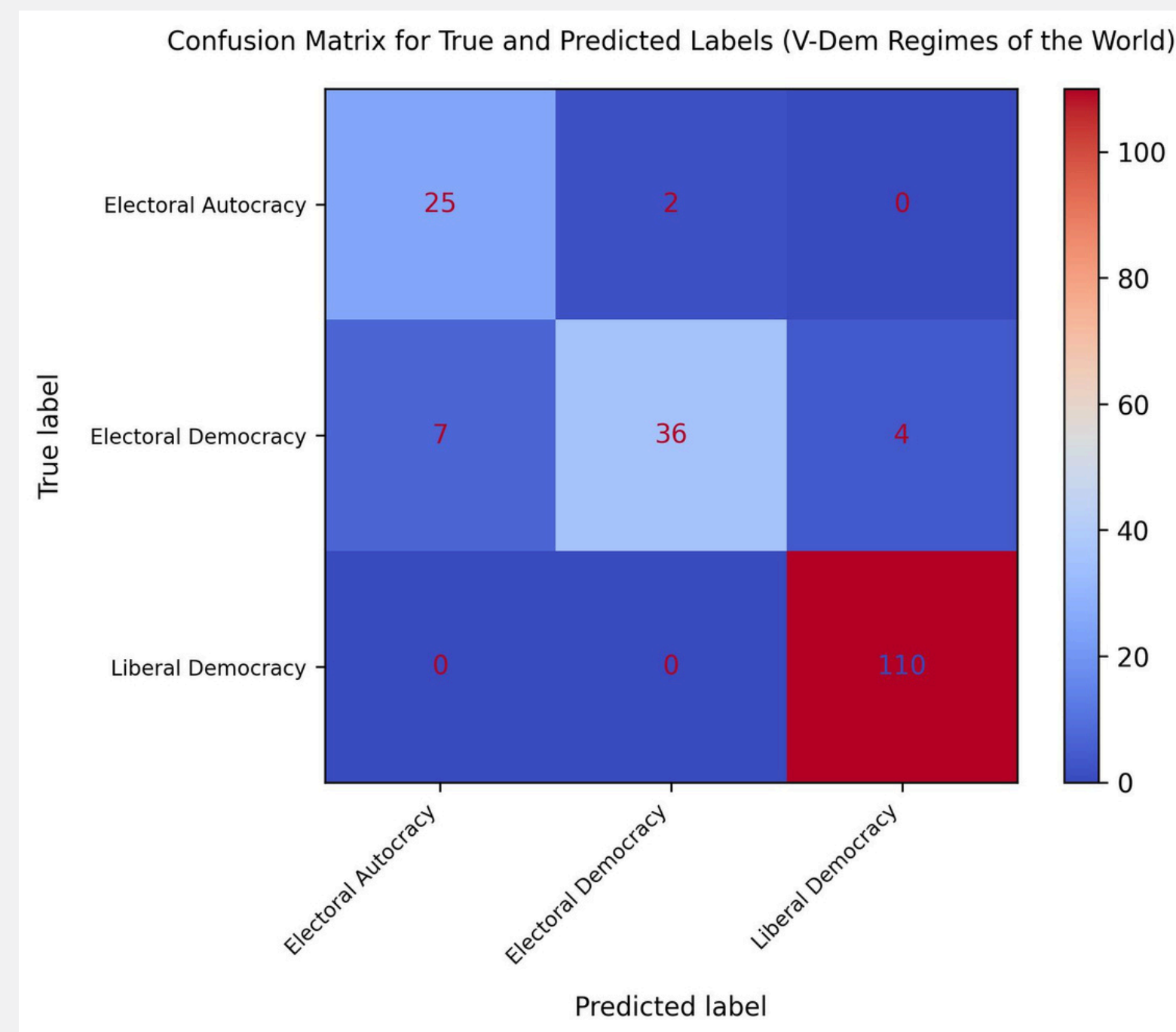
Results:

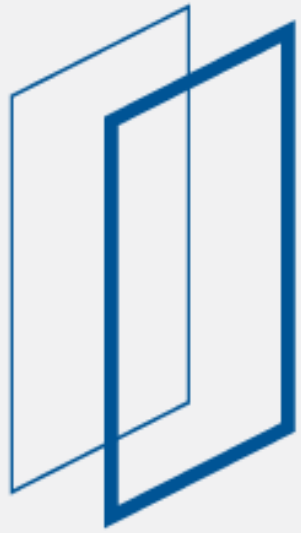
- Strong performance for identifying instances of liberal democracy

Considerations:

- Refinement to correctly identify instances of electoral democracy

	precision	recall	f1-score	support
elect_auto	0.78	0.93	0.85	27
elect_dem	0.95	0.77	0.85	47
lib_dem	0.96	1.00	0.98	110
accuracy			0.93	184
macro avg	0.90	0.90	0.89	184
weighted avg	0.93	0.93	0.93	184

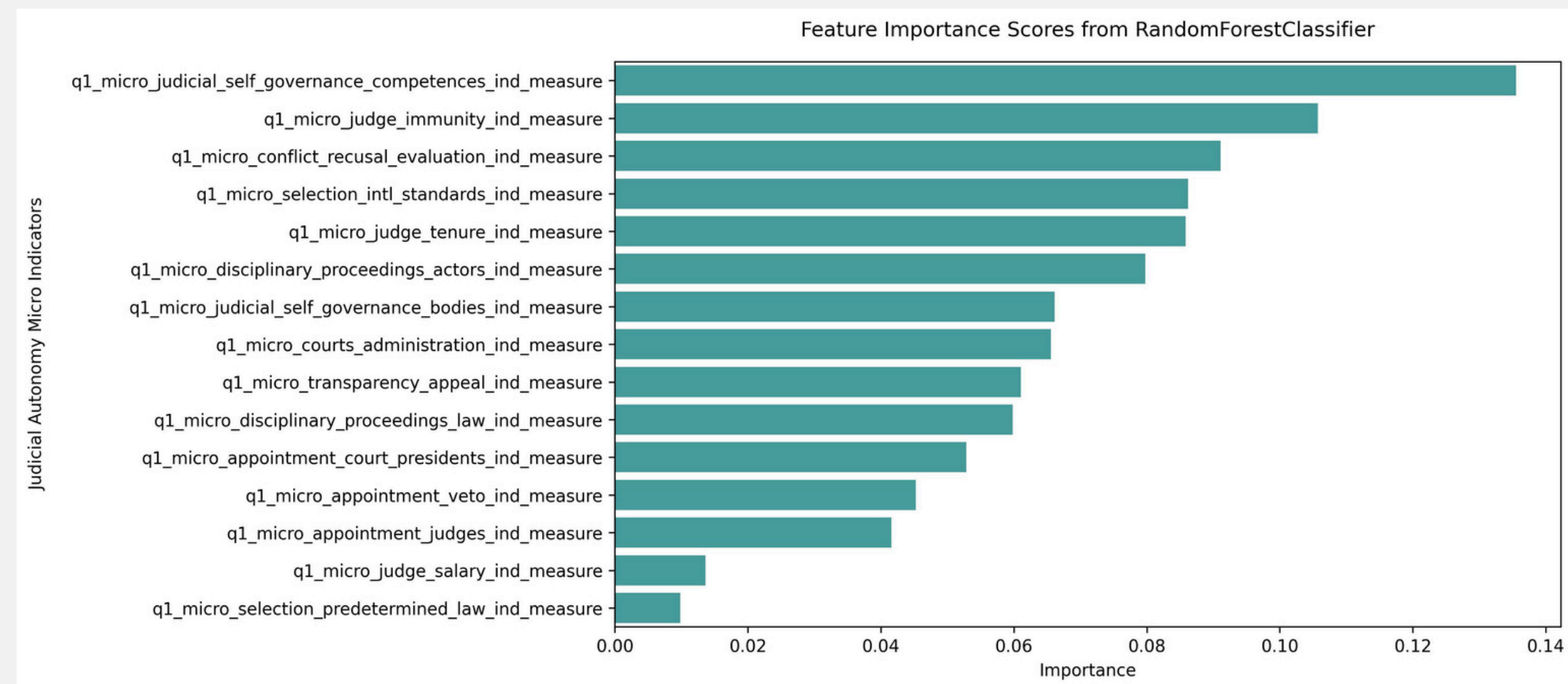


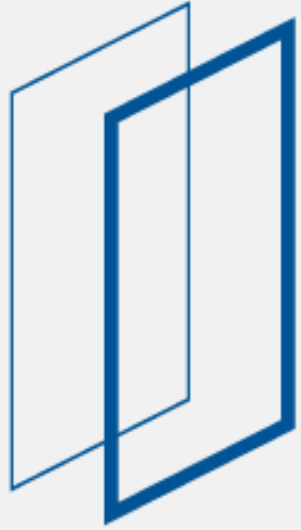


Random Forest Classifier

Results:

- Competences of the judicial self-governance bodies (i.e. bodies that manage and oversee the administration and internal matters of the judiciary) is the most predictive feature for regime type.





Random Forest Regressor

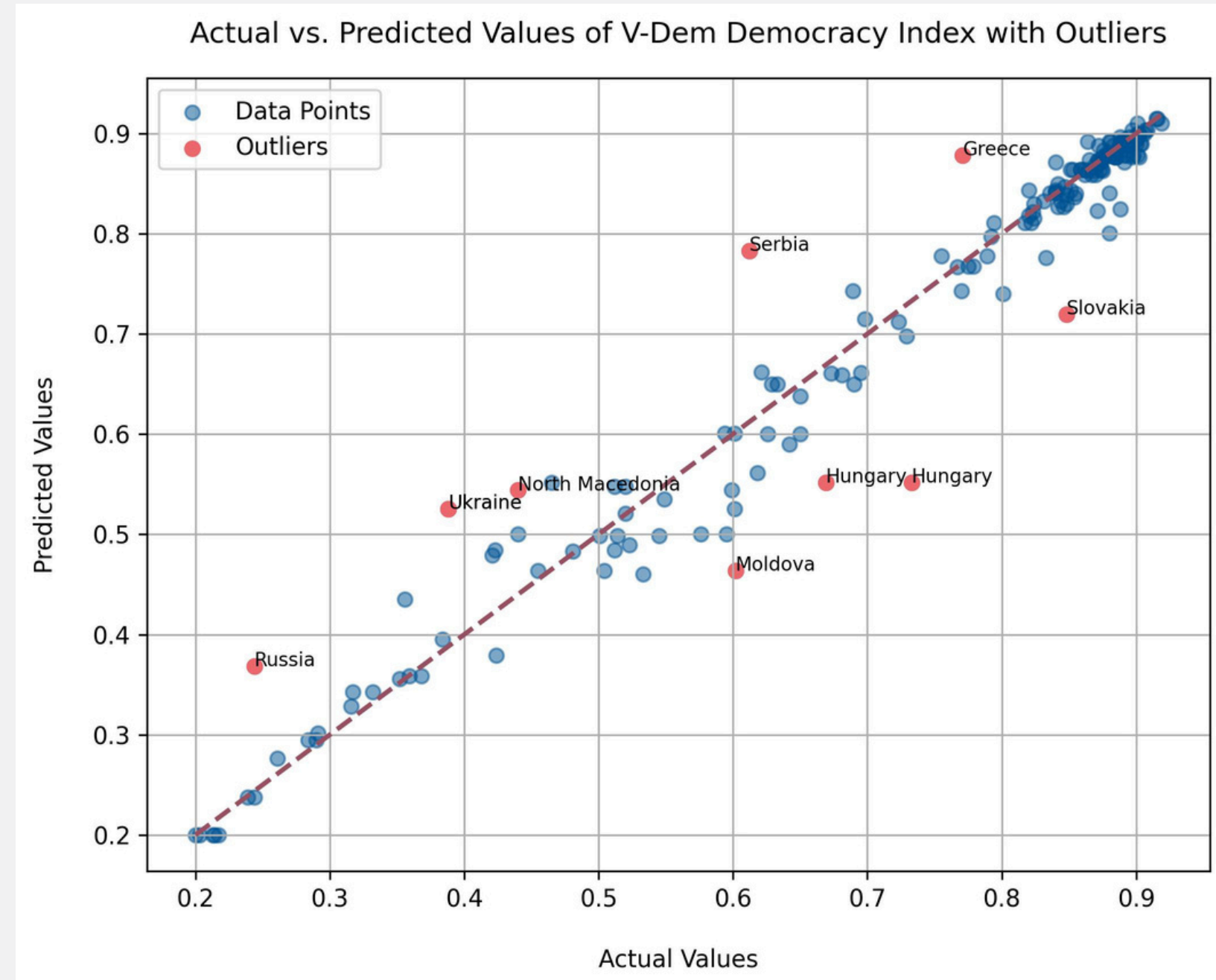
Results:

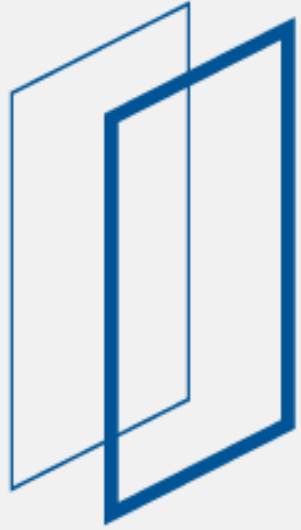
- Strong performance when predicting the democracy index measures
- High level of reliability

Considerations:

- Address potential overfitting
- Check generalizability

Mean Squared Error: 0.00
Root Mean Squared Error: 0.04
R² Score: 0.96

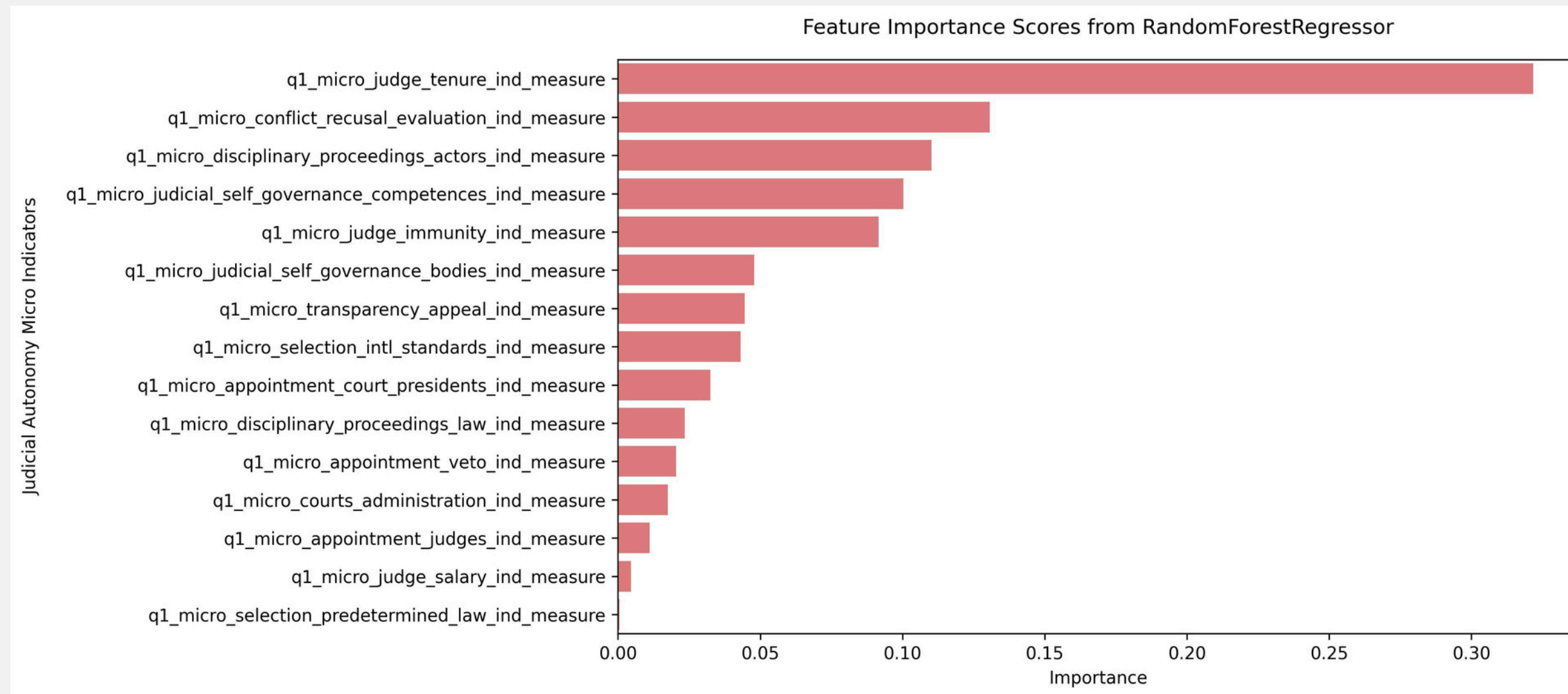


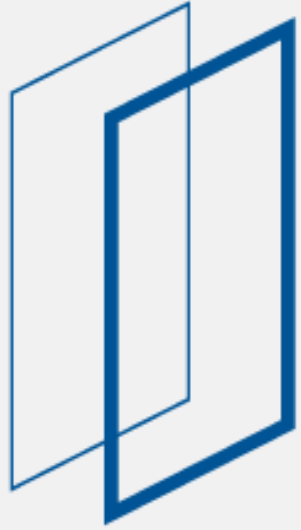


Random Forest Regressor

Results:

- Tenure or the term in office of judges is the most predictive feature for predicting democracy index measures.





Conclusion

Can written law predict the quality of democracy?

- Weak but recognizable correlation between the quality of judicial independence and the overall quality of democracy.
- Judicial independence can only be one crucial pillar among many.
- Challenges in recognizing the subtle erosion of democracy.

We cannot predict the (political) future, but we can prepare ourselves by:

- Identifying critical elements of judicial systems
- Identifying potential threats to judicial independence and democracy
- Monitoring incremental change
- Refining measures and approaches to capture "middle-ground" cases

Thank you!



Ironhack Data Analytics Final Project -- FINALLY DONE! 🙌